

Analytic Solution for Velocity Driven Flow in Power Law Model

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1 Derivation for Analytic Solution

To validate the power-law collision model, we compare against an analytic solution for a simple, well-characterised flow: steady, velocity-driven flow of a power-law fluid in a rigid channel.

Here we will derive the analytic solution for the velocity-driven flow case,

$$u_x(y) = U_{\max} \left[1 - \left(\frac{2|y - N_y/2|}{N_y} \right)^{(n+1)/n} \right]$$

the pressure drop required to observe a maximum velocity $U_{\max}$,

$$\Delta p = \frac{2mN_x}{N_y} \left(\frac{2(n+1)U_{\max}}{nN_y} \right)^n$$

and finally an expression for $U_{\max}$ in terms of the parameters m, n, ρ, Re and N_y ,

$$U_{\max} = \left[\frac{\rho}{m\text{Re}} N_y^{(2n-1)/n} \left(\frac{nN_y}{2(n+1)} \right)^{n-1} \right]^{1/(n-2)}$$

We consider a 2D velocity-driven flow in a channel of height N_y , with physical coordinates (x, y) . We impose four standard simplifying assumptions:

1. Steady flow ($\frac{\partial}{\partial t} = 0$).
2. Fully developed flow, i.e. no variation along the channel ($\frac{\partial}{\partial x} = 0$).
3. Unidirectional flow, i.e. no cross-stream velocity component ($\mathbf{u} = (u_x, 0)$).
4. Incompressible flow ($\rho = \text{const}$).

Continuity. Under these assumptions the continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_x)}{\partial x} + \frac{\partial(\rho u_y)}{\partial y} = 0,$$

is satisfied identically: every term vanishes by assumptions 1–4.

Momentum balance. In the x -direction, the momentum equation is

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} \right) + \rho g_x$$

The left-hand side vanishes entirely: the first term by steadiness, and the remaining two because $u_y = 0$ and $\frac{\partial u_x}{\partial x} = 0$. Similarly τ_{xx} , which depends only on $\frac{\partial u_x}{\partial x}$, does not contribute. This leaves the balance between the pressure gradient, the shear stress gradient,

$$\frac{\partial p}{\partial x} = \frac{\partial \tau_{yx}}{\partial y}$$

In the y -direction, the same argument (with $u_y = 0$ throughout) gives

$$\begin{aligned} \rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) &= -\frac{\partial p}{\partial y} + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \right) + \rho g_y \\ \Rightarrow \frac{\partial p}{\partial y} &= 0 \end{aligned}$$

which tells us that $p = p(x)$ only. Combining the two momentum balances, we are left with a single governing equation for the shear stress profile,

$$\frac{dp}{dx} = \frac{\partial \tau_{yx}}{\partial y} \tag{1}$$

The left-hand side of (1) is a function of x alone, while τ_{yx} in general depends on y ; for the equation to hold at every (x, y) , both sides must in fact equal the same constant.

Pressure gradient. Considering the left-hand side first,

$$\frac{dp}{dx} = c_0 \implies p(x) = c_0 x + c_1 \tag{2}$$

for constants c_0, c_1 . Let the pressure at the inlet and outlet be p_{in} and p_{out} respectively, that is, $p(0) = p_{in}$ and $p(N_x) = p_{out}$. Thus, $c_1 = p_{in}$ and $c_0 = -\Delta p/N_x$ where $\Delta p = p_{in} - p_{out}$ is the pressure drop, and we arrive at

$$p(x) = p_0 - \frac{\Delta p}{N_x} x \tag{3}$$

Shear stress profile. With $c_0 = -\Delta p/N_x$, equation (1) becomes an ODE for the shear stress alone,

$$\frac{\partial \tau_{yx}}{\partial y} = -\frac{\Delta p}{N_x} \implies \tau_{yx}(y) = -\frac{\Delta p}{N_x}y + c_2 \quad (4)$$

for some constant c_2 . The flow is symmetric about the channel centreline $y = N_y/2$, so by symmetry the shear stress must vanish exactly at the centreline. Imposing $\tau_{yx}(N_y/2) = 0$ fixes $c_2 = N_y\Delta p/(2N_x)$, giving the linear shear stress profile

$$\tau_{yx}(y) = \frac{\Delta p}{N_x} \left(\frac{N_y}{2} - y \right). \quad (5)$$

Note that this result is entirely independent of the constitutive law of the fluid: it follows purely from the momentum balance and symmetry, and holds for Newtonian and power-law fluids alike. The rheology only enters once we relate τ_{yx} to the velocity gradient. At this step we could in future use a different constitutive law other than the power law.

Constitutive law and velocity profile. We now introduce the power-law constitutive relation,

$$\tau_{yx} = m \left| \frac{du_x}{dy} \right|^{n-1} \frac{du_x}{dy}, \quad (6)$$

From (5), τ_{yx} is positive for $y < N_y/2$ and negative for $y > N_y/2$; by (6), du_x/dy must have the same sign, meaning the velocity increases towards the centreline from both walls, as expected for a driven channel flow.

We will solve the problem on the lower half of the channel, $y < N_y/2$, where all quantities are positive, and use the symmetry of the profile to extend the result to the full domain $y \in [0, N_y]$ afterwards.

For $y < N_y/2$, both sides of (6) are positive, so the absolute value can be dropped and we can equate (5) with (6) directly,

$$m \left(\frac{du_x}{dy} \right)^n = \frac{\Delta p}{N_x} \left(\frac{N_y}{2} - y \right) \implies \frac{du_x}{dy} = \left[\frac{\Delta p}{mN_x} \left(\frac{N_y}{2} - y \right) \right]^{1/n}. \quad (7)$$

Integrating with respect to y gives the velocity profile up to a constant,

$$u_x(y) = -\frac{n}{n+1} \left(\frac{\Delta p}{mN_x} \right)^{1/n} \left(\frac{N_y}{2} - y \right)^{(n+1)/n} + c_3. \quad (8)$$

We fix c_3 using the no-slip condition at the lower wall, $u_x(0) = 0$, which gives

$$c_3 = \frac{n}{n+1} \left(\frac{\Delta p}{mN_x} \right)^{1/n} \left(\frac{N_y}{2} \right)^{(n+1)/n},$$

and substituting back into (8) yields the velocity profile on the lower half of the channel,

$$u_x(y) = \frac{n}{n+1} \left(\frac{\Delta p}{mN_y} \right)^{1/n} \left[\left(\frac{N_y}{2} \right)^{(n+1)/n} - \left(\frac{N_y}{2} - y \right)^{(n+1)/n} \right]. \quad (9)$$

Full profile. By the symmetry of the problem about $y = N_y/2$, the solution on the upper half of the channel is obtained simply by replacing $(N_y/2 - y)$ with $(y - N_y/2)$. Combining both halves into a single expression valid across the whole channel $y \in [0, N_y]$, we arrive at the analytic velocity profile for force-driven channel flow of a power-law fluid,

$$u_x(y) = \frac{n}{n+1} \left(\frac{\Delta p}{m N_y} \right)^{1/n} \left[\left(\frac{N_y}{2} \right)^{(n+1)/n} - \left| \frac{N_y}{2} - y \right|^{(n+1)/n} \right] \quad (10)$$

We can introduce the maximum velocity $U_{\max}$ as the value of the velocity at the centreline $y = N_y/2$,

$$U_{\max} = \frac{n}{n+1} \left(\frac{\Delta p}{m N_x} \right)^{1/n} \left(\frac{N_y}{2} \right)^{(n+1)/n} \quad (11)$$

to write our expression in (10) as,

$$u_x(y) = U_{\max} \left[1 - \left(\frac{2|y - N_y/2|}{N_y} \right)^{(n+1)/n} \right] \quad (12)$$

Pressure drop condition. Moreover, we can use (11) to find an expression for the required pressure drop Δp to observe a maximum velocity $U_{\max}$,

$$\Delta p = \frac{2m N_x}{N_y} \left(\frac{2(n+1)U_{\max}}{n N_y} \right)^n \quad (13)$$

Kinematic viscosity. One can find the expression for the kinematic velocity at the centreline as

$$\nu_c = \frac{m}{\rho} \left(\frac{\Delta p}{2m N_x} \right)^{(n-1)/n} \quad (14)$$

using the equation $\tau(y) = \frac{1}{2} + \frac{3}{\rho} \mu_{\text{app}}$ and $\mu_{\text{app}} = m \left| \frac{du_x}{dy} \right|^{n-1}$ together with the analytic solution for the velocity $u_x(y)$, and then using the relation

$$\nu_c = \frac{1}{3} \left(\tau_c - \frac{1}{2} \right)$$

where we have that

$$\tau_c := \tau \left(\frac{N_y}{2} - \frac{1}{2} \right) = \frac{1}{2} + \frac{3m}{\rho} \left(\frac{\rho g}{2m} \right)^{(n-1)/n}$$

Maximum velocity. Using this and (13), we can find an expression for the kinematic velocity in terms of the maximum velocity,

$$\nu_c = \frac{m}{\rho} \left(\frac{1}{N_y} \right)^{(n-1)/n} \left(\frac{2(n+1)U_{\max}}{n N_y} \right)^{n-1} \quad (15)$$

which we use with our definition of the Reynolds number for the power-law,

$$\text{Re} = \frac{U_{\max} N_y}{\nu_c} \quad (16)$$

to find an expression for the maximum velocity,

$$U_{\max} = \left[\frac{\rho}{m\text{Re}} N_y^{(2n-1)/n} \left(\frac{nN_y}{2(n+1)} \right)^{n-1} \right]^{1/(n-2)} \quad (17)$$